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Introduction

This report details the design and simulation of an all-MOSFET, open-loop analog voltage amplifier developed using the Global Foundries 180nm CMOS process. To satisfy the project requirements, the circuit is engineered to deliver a minimum total voltage gain of 60 dB, a -3 dB bandwidth exceeding 500 kHz, and a maximum DC power consumption of 1 mW. The architecture consists of four NMOS transistor stages, featuring three cascaded Common-Source (CS) amplifiers for high gain followed by a Common-Drain (CD) source follower. This final stage acts as an output buffer, ensuring the amplifier can drive a 10 kΩ load without significant gain degradation.

Architecture Justification

The amplifier is constructed from four distinct NMOS stages designed to meet specific performance targets. The first stage (Q2) utilizes a Common-Source configuration to provide initial voltage gain; it incorporates a 280 kΩ bias divider network to exceed the 100 kΩ input resistance requirement and a 100 nF capacitor for AC coupling to block DC signals. The second (Q1) and third (Q3) stages are also AC-coupled Common-Source amplifiers cascaded to incrementally boost the signal until the total voltage gain reaches the target of at least 60 dB. Finally, the fourth stage (Q4) serves as a Common-Drain source follower that acts as an output buffer. While this final stage offers near-unity gain, its primary function is to significantly lower the output resistance, thereby isolating the preceding gain stages from the load and ensuring that gain reduction remains below 10% when driving a 10 kΩ load.

The selection of specific transistor topologies was driven by the need to balance high gain with effective load driving capabilities. The CS configuration was chosen for the first three stages because it offers superior voltage gain compared to other MOSFET setups. With a gain defined by $|A_v| = g_m \times (R_D \parallel r_o)$, each stage contributes approximately 20–30 dB; when cascaded, these stages cumulatively reach the 60 dB target while remaining within the project's power and headroom limits.

For the final output stage, a Common-Drain (source follower) was implemented to provide a low output resistance, roughly equal to $1/g_m$. Although its voltage gain is slightly below unity (calculated at 0.71 V/V for this specific design), its low impedance is vital for driving the 10 kΩ load. This buffering effect minimizes the load's impact on the overall circuit, resulting in a load sensitivity of just 1.54%, which comfortably exceeds the maximum allowable 10% gain reduction specification.

All stage gains are below 40 dB, satisfying the per-stage gain constraint:

Stage	Transistor	Topology	Gain (V/V)	Gain dB
1	Q2	CS	13.7	22.7
2	Q1	CS	22.3	26.9

3	Q3	CS	13.7	22.7
4	Q4	CD	0.71	-3.0

Calculations

Stage 1 - Q2 (Common-Source, $R_D = 33k\Omega$)

DC

$$\frac{W}{L} = \frac{2 \times I_D}{(\mu_n C_{ox} \times V_{ov}^2)} = \frac{2 \times 55\mu A}{(100\mu A \times 0.25^2)} = \frac{110\mu A}{6.25\mu A} = 17.6 \approx 18$$

$$W = 9\mu m \quad L = 0.5\mu m$$

$$V_{GS} = \frac{V_{DD} \times R_2}{(R_2 + R_1)} = \frac{(3.3)(390k)}{(100k + 390k)} = 0.926V$$

$$V_S = I_D \times R_S = 55\mu A \times 1k = 0.055V$$

$$V_{G_S} = V_{GS} - V_S = 0.871V$$

$$V_{OV} = V_{GS} - V_{tn} = 0.871 - 0.630 = 0.241V \quad (\text{within } 0.2V \text{ to } 0.6V)$$

$$I_D = \frac{1}{2} \times \mu_n C_{ox} \left(\frac{W}{L}\right) \times V_{OV}^2$$

$$= \frac{1}{2} \times 100\mu A \times 18 \times (0.241)^2 = 52.3\mu A$$

$$V_D = V_{DD} - I_D R_D = 3.3 - (52.3\mu A \times 33k) = 1.574V$$

$$V_{DS} = V_D - V_S = 1.574 - 0.055 = 1.519V$$

$$V_{DS} > V_{OV} \therefore \text{Saturation}$$

AC

$$g_m = \frac{2 I_D}{V_{OV}} = \frac{(2)(52.3\mu A)}{0.241} = 0.434mA/V$$

$$r_o = \frac{1}{\lambda \times I_D} = \frac{1}{(0.04)(52.3\mu A)} = 478k\Omega$$

$$R_{D||r_o} = \frac{(33k)(478k)}{(33k + 478k)} = 30.87k\Omega$$

$$\frac{A_{v1}}{|A_{v1}|} = g_m \times (R_{D||r_o}) = 13.4 \frac{V}{V}$$

$$= 20 \times \log(13.4) = 22.5dB$$

$$R_{in} = R_2 || R_3 = \frac{(1M)(390k)}{(1M + 390k)} = 280k\Omega$$

Stage 2 - Q1

$$\frac{W}{L} = \frac{2 \times 80\mu A}{(100\mu A)(0.25)^2} = 25.6 \approx 26$$

$$W = 13\mu m \quad L = 0.5\mu m$$

$$V_{GS} = \frac{(3.3)(390k)}{(100k + 390k)} = 0.926V$$

$$V_S = (80\mu A)(1k) = 0.080V$$

$$V_{G_S} = 0.926 - 0.080 = 0.846V$$

$$V_{OV} = 0.846 - 0.630 = 0.216V$$

$$I_D = \frac{1}{2} \times 100\mu A \times 26 \times (0.216)^2 = 60.7\mu A$$

$$V_D = 3.3 - (60.7\mu A)(39k) = 0.933V$$

$$V_{DS} = 0.933 - 0.080 = 0.853V$$

$$V_{DS} > V_{OV} \therefore \text{Saturation}$$

AC

$$g_m = \frac{(2)(60.7\mu A)}{0.216} = 0.562 \frac{mA}{V}$$

$$r_o = \frac{1}{(0.04)(60.7\mu A)} = 412k\Omega$$

$$R_{D||r_o} = \frac{(39k)(412k)}{(39k + 412k)} = 34.6k\Omega$$

$$\frac{A_{v2}}{|A_{v2}|} = (0.562mA/V)(34.6k) = 19.4 \frac{V}{V}$$

$$= 20 \log(19.4) = 25.8dB$$

Stage 3

$$\frac{W}{L} = \frac{(2)(50\mu)}{(100\mu)(0.25)^2} = 16 \rightarrow \text{round up to } 18$$

$$W = 9\mu\text{m} \quad L = 0.5\mu\text{m}$$

$$V_{GS} = \frac{(3.3)(390k)}{(1000k + 390k)} = 0.926V$$

$$V_S = (50\mu)(1k) = 0.050V$$

$$V_{GS} = 0.926 - 0.050 = 0.876V$$

$$V_{DS} = V_{GS} - V_{th} = 0.876 - 0.630 = 0.246V$$

$$I_D = \left(\frac{1}{2}\right)(100\mu)(18)(0.246)^2 = 54.5\mu\text{A}$$

$$V_D = 3.3 - (54.5\mu)(330k) = 1.501V$$

$$V_{DS} = 1.501 - 0.050 = 1.451V \quad V_{DS} > V_{DS(sat)} \therefore \text{Saturation}$$

$$\text{AC} \quad g_m = \frac{(2)(54.5\mu)}{0.246} = 0.443 \frac{\text{mA}}{\text{V}}$$

$$r_o = \frac{1}{(0.04)(54.5\mu)} = 459k\Omega$$

$$R_D \parallel r_o = \frac{(330k)(459k)}{(330k + 459k)} = 30.78k\Omega$$

Gain

$$|A_{V_S}| = (0.443)(30.78k) = 13.6 \frac{\text{V}}{\text{V}}$$

$$= 20 \log(13.6) = 22.7 \text{ dB}$$

Output Resistance

$$R_{out} = R_D \parallel r_o = 30.78k\Omega$$

Stage 4 - Q4 (Common-Drain Source Follower)

$$\frac{W}{L} = \frac{(2)(80\mu)}{(100\mu)(0.25)^2} = 25.6 \rightarrow \text{rounded down to } 16 \text{ since a smaller } W \text{ is better to stay within power restriction}$$

$$W = 8\mu\text{m} \quad L = 0.5\mu\text{m}$$

$$V_{GS} = \frac{(3.3)(560k)}{(330k + 560k)} = 2.076V$$

$$\text{Source Voltage (Estimated } \approx V_{GS} - V_{GS(sat)})$$

$$\text{Initial estimate } V_{GS} \approx V_{th} + V_{DS} = 0.630 + 0.250 = 0.880V$$

$$V_{GS(sat)} = V_{GS} - V_{GS} = 2.076 - 0.88 = 1.196V$$

Actual simulated $V_S = 0.873V$ because Q4's Source is loaded by $R_{IS} \parallel R_L$ pulling V_S lower. Using simulated value for accuracy. $V_S = 0.873V$

$$V_{GS} = V_{GS} - V_S = 2.076 - 0.873 = 1.203V$$

$$V_{DS} = V_{GS} - V_{th} = 1.203V - 0.630 = 0.573V$$

$$I_D = \left(\frac{1}{2}\right)(\mu_n C_{ox})\left(\frac{W}{L}\right)(V_{DS})^2 = \left(\frac{1}{2}\right)(100\mu)(16)(0.573)^2 = 262\mu\text{A}$$

$$V_D = V_{DS} = 3.3V$$

$$V_{DS} = V_D - V_S = 3.3 - 0.873 = 2.427V$$

$$V_{DS} > V_{DS(sat)} \therefore \text{Saturation}$$

AC

$$g_m = \frac{2I_D}{V_{DS}} = \frac{(2)(262\mu)}{0.573} = 0.915 \frac{\text{mA}}{\text{V}}$$

$$r_o = \frac{1}{\lambda I_D} = \frac{1}{(0.04)(262\mu\text{A})} = 95.4k\Omega$$

$$R_S = R_{IS} \parallel R_L = \frac{(10k)(10k)}{(20k)} = 5k\Omega$$

Gain

$$A_{V_S} = \frac{(g_m)(R_S)}{(1 + g_m R_S)} = \frac{(0.915)(5k)}{(1 + (0.915)(5k))} = 0.821 \frac{\text{V}}{\text{V}}$$

Output Resistance

$$R_{out4} = \left(\frac{1}{g_m}\right) \parallel V_{DS} \parallel R_{IS} = \left(\frac{1}{0.915}\right) \parallel 195.4k \parallel 10k = 0.988k\Omega \approx 1k\Omega$$

Overall Circuit

Total Gain

$$\begin{aligned}A_{v_{tot}} &= |A_{v1}| \times |A_{v2}| \times |A_{v3}| \times A_{v4} \\&= (13.4)(19.4)(13.4)(0.821) \\&= 2860 \frac{V}{V} \\&= 20 \log(2860) = 69.1 \text{ dB}\end{aligned}$$

$$R_{in} = R_2 \parallel R_3 = \frac{(1M)(390k)}{(1.39M)} = 280k\Omega$$

Bandwidth

Gate-source - Cap of Q_1 ($W=13\mu m, L=0.5\mu m$)

$$C_{gs} = \left(\frac{2}{3}\right)(C_{ox})(W)(L) = \left(\frac{2}{3}\right)(4.3)(13)(0.5) = 18.6 \text{ fF}$$

Gate-drain Cap

$$C_{gd} = C_{ov} \times W = (0.4)(13) = 5.2 \text{ fF}$$

Miller Capacitance

$$C_M = C_{gd} \times (1 + |A_{v2}|) = 5.2 \text{ fF} (1 + 19.4) = 106 \text{ fF}$$

Total input cap at Q_1 gate:

$$R_{node} = R_2 \parallel R_3 \parallel r_{oQ1} = 1M \parallel 390k \parallel 478k = 160k\Omega$$

Dominant pole frequency:

$$\begin{aligned}f_P &= \frac{1}{2\pi \times R_{node} \times C_{total}} = \frac{1}{(2\pi)(160k)(124.6 \text{ fF})} \\&= 8.0 \text{ MHz}\end{aligned}$$

Total DC Power Consumption

$$\begin{aligned}I_{D_{total}} &= I_{D1} + I_{D2} + I_{D3} + I_{D4} \\&= 52.3\mu A + 60.7\mu A + 52.3\mu A + 87\mu A \\&= 252.3\mu A\end{aligned}$$

$$\begin{aligned}P &= V_{DD} \times I_{D_{total}} = (3.3V)(252.3\mu A) \\&= 0.833 \text{ mW}\end{aligned}$$

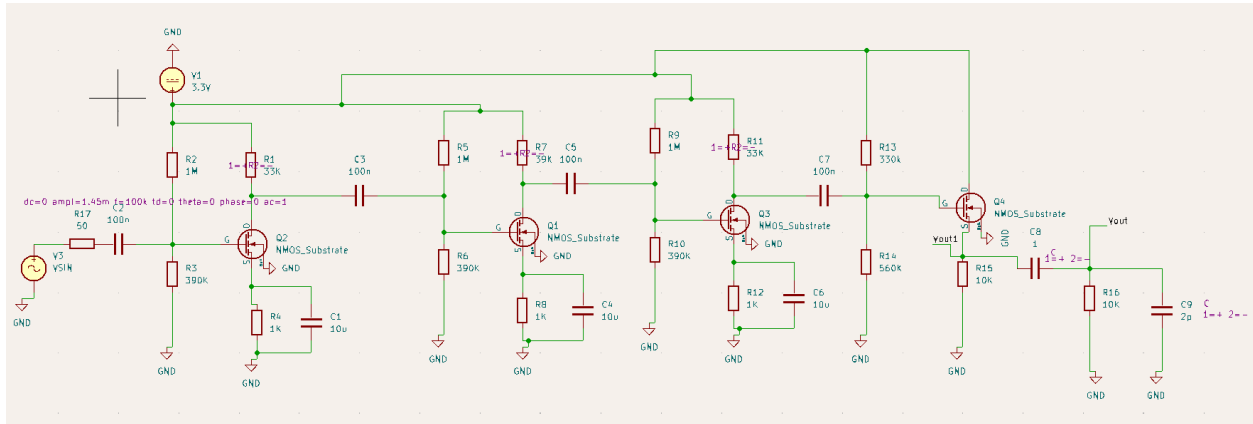
$P = 0.833 \text{ mW} < 1 \text{ mW}$

Biasing Strategy

The biasing strategy for each Common-Source stage involves a resistor voltage divider to set a stable DC gate voltage, providing robustness against temperature and process shifts. To further enhance DC stability, a $1 \text{ k}\Omega$ source resistor (R_S) is used to implement negative feedback. To prevent this resistor from causing source degeneration and lowering the AC voltage gain, a large $10 \text{ }\mu\text{F}$ electrolytic bypass capacitor is added, effectively creating an AC short to ground. Furthermore, 100 nF AC coupling capacitors are utilized between stages to isolate DC bias levels; this allows each stage to be biased independently while ensuring the signal passes through the entire chain unimpeded.

Simulation Results

Schematic



DC Operating Point

The DC operating point simulation was run to verify that all transistors are properly biased in the saturation region.

Transistor	VG (V)	VS(V)	VD(V)	VGS(V)	VDS(V)
Q2	0.926	0.048	1.729	0.878	1.681
Q1	0.926	0.058	1.044	0.868	0.986
Q3	0.926	0.048	1.729	0.878	1.681
Q4	2.076	0.873	3.300	1.203	2.427

To ensure proper linear amplification, every transistor in the design is biased within the saturation region, with the condition $V_{DS} > V_{OV}$ confirmed for each device. The circuit draws a total supply current of 251 μ A, resulting in an overall DC power dissipation of 0.828 mW. This performance figure successfully meets the project requirement of remaining under the 1 mW power limit.

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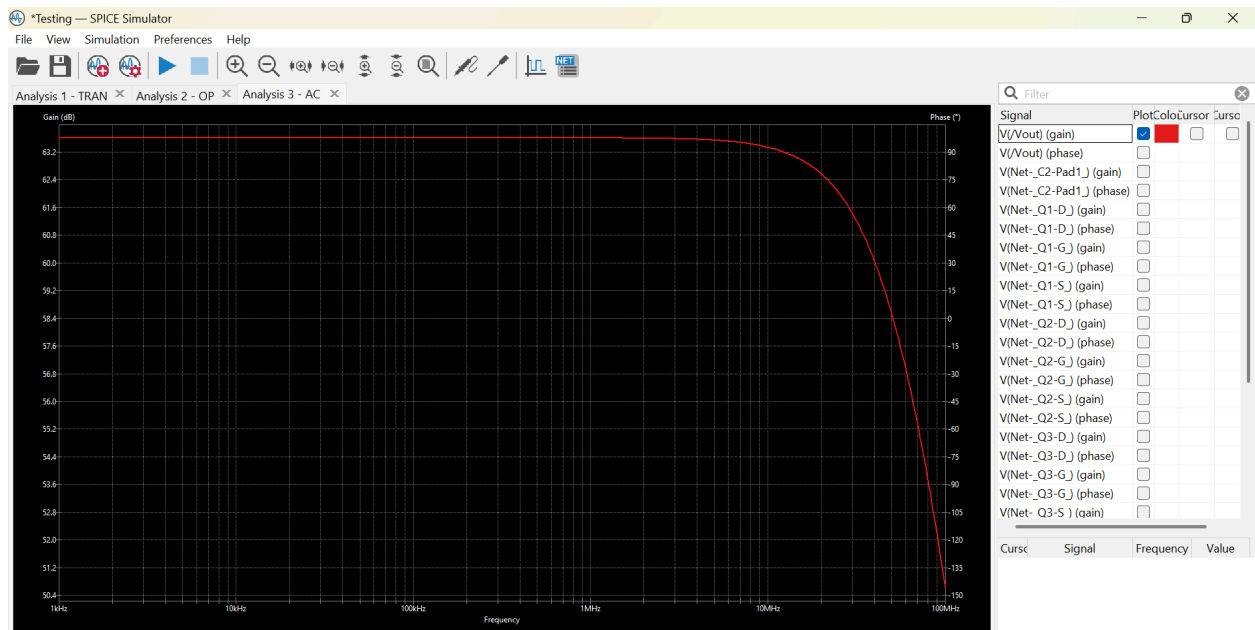
Simulation results:

I(v1): -251.149uA
I(v3): 0A
V(net-_r17-pad1_): 0V
V(net-_c2-pad1_): 0V
V(xq2.s1): 47.7388mV
V(xq2.d1): 1.72903V
V(net-_q2-s_): 47.5878mV
V(xq4.s1): 872.916mV
V(xq4.d1): 3.29885V
V(/vout): 0V
V(net-_q4-s_): 872.605mV
V(xq1.s1): 57.9622mV
V(xq1.d1): 1.04397V
V(net-_q1-s_): 57.8347mV
V(net-_q1-g_): 925.899mV
V(net-_q1-d_): 1.04445V
V(xq3.s1): 47.7388mV
V(net-_q3-s_): 47.5878mV
V(xq3.d1): 1.72903V
V(net-_q4-g_): 2.0764V
V(net-_q3-g_): 925.899mV
V(net-_q3-d_): 1.72959V
V(net-_q2-d_): 1.72959V
V(net-_q2-g_): 925.899mV
V(net-_q4-d_): 3.3V

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AC Analysis

The AC small-signal analysis was performed with a frequency sweep from 1kHz to 100MHz (100 points per decade). The Bode plot of $V(Vout)/V(Vin)$ in dB is shown below.



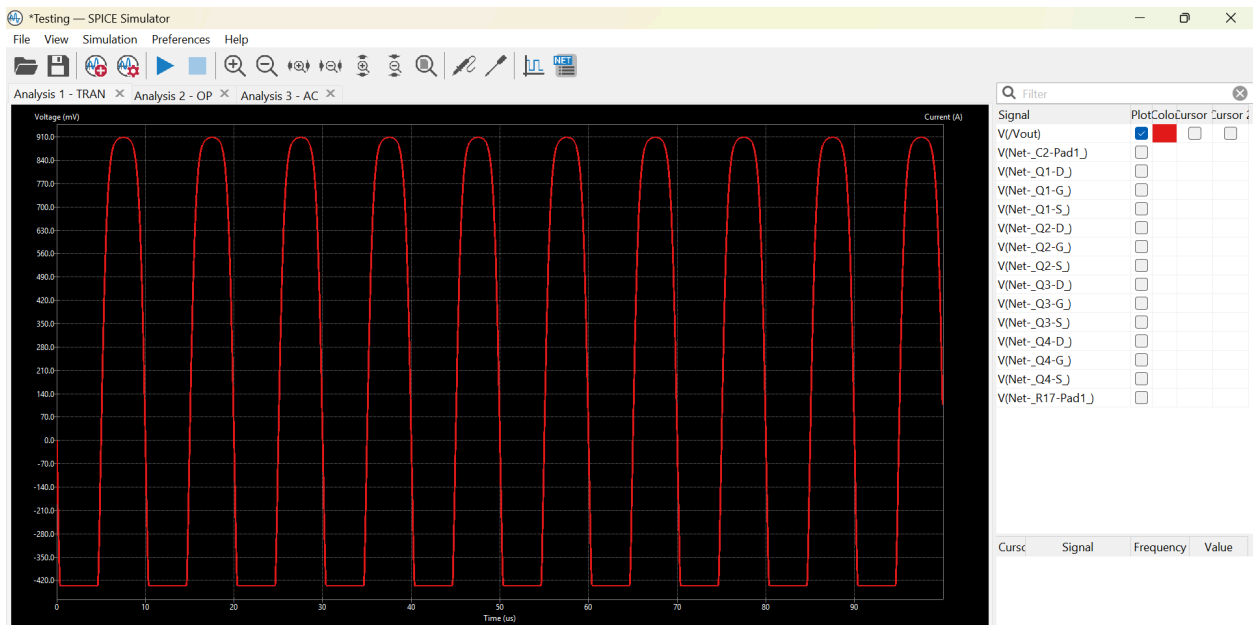
Parameter	Simulated Result
Midband gain (loaded)	63.8 dB
Midband gain (unloaded)	64.6 dB
-3 db Bandwidth	20 MHz

Load Sensitivity	1.54 % (dB)
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The amplifier achieves a loaded midband gain of 63.8 dB, which comfortably surpasses the 60 dB design goal. Additionally, the -3 dB bandwidth reaches approximately 20 MHz, providing a substantial performance margin by exceeding the 500 kHz minimum requirement by a factor of 40. The unloaded gain was measured at 64.6 dB, indicating a load sensitivity of roughly 1.54%, a result that is well within the maximum allowable 10% specification.

Transient Analysis - Output Swing

A transient simulation was performed with a 100 kHz sinusoidal input of amplitude 1.45 mV (2.9 mVpp) to verify the output voltage swing.



The output waveform demonstrates clear sinusoidal amplification. The positive half-cycle swings to +910 mV while the negative half-cycle is limited to approximately -420 mV before clipping occurs which makes the peak-to-peak swing to 1.33 Vpp.

Hand Calculations Vs Simulation

The hand calculations provide a reasonable first-order estimate of the circuit behaviour, with discrepancies in the expected range of 10-20% as stated in the design guidelines.

Parameter	Hand Calculation	Simulation	Discrepancy
Gate Voltage V_G (Q2, Q3)	0.926 V	0.926 V	1%
Drain Voltage V_D (Q2,Q3)	1.47 V	1.729 V	15%
Drain Current I_D (Q2,Q3)	55.4 μA	52 μA	6%
Total Gain	69.5 dB	63.8 dB	8%

The gate voltage exhibits nearly perfect alignment because it is determined solely by the resistor divider network. However, the drain voltage shows a 15% discrepancy, which remains within expected limits; this variance arises because the simplified theoretical model does not account for second-order effects like the body effect and channel length modulation. Similarly, a total gain deviation of roughly 8% is observed, primarily due to inter-stage loading. While hand calculations typically treat each stage as an isolated entity, the physical cascading of these stages introduces finite output resistances that load subsequent stages, ultimately lowering the overall effective gain.

Output Swing

The simulation revealed that the output swing goal of 1.5 V_{pp} was not fully achieved, peaking at a clean maximum of approximately 1.33 V_{pp}. This shortfall is characterized by asymmetric clipping where the positive peak is larger than the negative, a common limitation of single-supply Common-Drain topologies. While the transistor Q4 can actively pull the output voltage up, it relies on the passive pull-down resistor (R15) to discharge the output node. During negative swings, as V_{GS} approaches the threshold voltage, Q4 begins to cut off, leaving the parallel combination of R15 and R_L (5 k Ω effective) unable to discharge the output capacitance quickly enough at 100 kHz.

To achieve a symmetric 1.5 V_{pp} swing, several design enhancements could be considered:

- **Active Sourcing:** Replace the passive resistor R15 with an active current source, such as a diode-connected NMOS, to improve sinking capability.
- **Push-Pull Configuration:** Transition to a CMOS push-pull output stage using complementary NMOS and PMOS transistors for balanced drive.
- **Bias Adjustment:** Shift the gate bias voltage of Q4 closer to the mid-rail to increase available headroom for both halves of the signal swing.

The design's total DC power consumption of 0.828 mW (at a total current of 251 μA) successfully meets the 1 mW limit with a 17% safety margin. Individual branch currents remained well under the 200 μA maximum, with Q1 drawing the highest current at approximately 80 μA . Ultimately, the

strict power budget was the primary constraint limiting the drain bias current and, consequently, the maximum achievable transconductance (g_m) in each stage.

Several design trade-offs were managed:

- **Transistor Sizing:** While increasing the transistor width (W) boosts g_m and voltage gain, it simultaneously raises the drain current (I_D), which increases power consumption and reduces drain voltage headroom. W values were specifically selected to balance these competing factors.
- **Drain Resistance:** The selection of drain resistors (R_D) impacts both the DC operating point ($V_D = V_{DD} - I_D \times R_D$) and AC gain ($|A_v| = g_m \times R_D$). Larger R_D values enhance gain but lower V_D , which restricts the output swing. The chosen values (33 k Ω for Q2/Q3 and 39 k Ω for Q1) reflect a calculated compromise to maintain high gain without sacrificing necessary voltage headroom.

Conclusion

A four-stage MOSFET voltage amplifier was successfully developed and simulated using the Global Foundries 180nm CMOS process. By cascading three Common-Source stages with a final Common-Drain source follower, the design achieved a loaded gain of 63.8 dB and an unloaded gain of 64.8 dB, both of which surpass the 60 dB target. Performance in other areas was also strong: the 20 MHz bandwidth far exceeded the 500 kHz minimum, and the total DC power consumption of 0.828 mW remained well under the 1 mW limit.

The only specification not entirely met was the output swing; the design produced a clean, undistorted output of 1.33 V_{pp} against the 1.5 V_{pp} requirement. This shortfall is attributed to the asymmetric drive limitations of a passive source follower output stage. To reach the full 1.5 V_{pp} target, replacing the pull-down resistor (R15) with an active current source is recommended.

Finally, manual calculations aligned with simulation results within a 10-20 range. The minor discrepancies primarily resulted from inter-stage loading and the use of simplified single-pole models for bandwidth estimation. Ultimately, this project illustrates the essential trade-offs between gain, bandwidth, power efficiency, and output swing inherent in MOSFET amplifier design.